

1. (a) $A'B' + A'D' + AC'B + AC'D'$
 (b) $XWY + WY + WYZ + W'ZX + W'ZY + W'$
 2. (a) $(P+R')(Q+R')(P+Q')(R+Q')$
 (b) ?
 3. $ABC + ABD$
 4. No. 真值表
 5. (a) $AB + BC \oplus BD$
 (b) ?

Digital Logic Design Homework 2

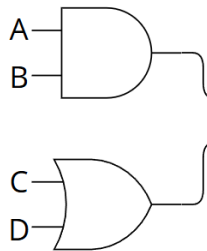
Instructions: Solve the following problems. Show all your steps clearly.

1. **Product of Sums:** Multiply out to obtain a sum of products. Simplify where possible.
- (a) $(A+B'+D')(A'+C')(A'+B+D')$
 $= A'B'D' + A'CD' + AC'B + AC'D'$
 $= XWY + WY + WYZ + W'ZX + W'ZY + W'$
- (b) $(W'+Y)(W+Z)(X+Y+Z)$
 $= (WY + W'Z)(X+Y+Z)$
 $= XWY + WY + WYZ + W'ZX + W'ZY + W'$

2. **Sum of Products:** Factor to obtain a product of sums. Simplify where possible.

- (a) $PQR + Q'R' + P'Q'R + QR$
 $= (P+R')(Q+R')(P+Q')(R+Q')$
 $= PQR + Q'R' + P'Q'R + QR$
 $= PQR + Q'(R'+Q') + QR$
 $= PQR + Q'(R'+Q') + QR$
 $= PQR + Q'R' + Q' = PQR + (R'+Q')$

3. **Circuit Simplification:** Write an expression for F and simplify.



$$F = AB(C+D) = ABC + ABD$$

$$x \oplus y = (x \equiv y)' = (x \leftrightarrow y)'$$

$$= [(x \rightarrow y) \wedge (y \rightarrow x)]'$$

$$= [(x \vee y) \wedge (y \vee x)]'$$

$$= (x \vee y)' \vee (y \vee x)'$$

$$= (x' \wedge y') \vee (y' \wedge x')$$

$$= x'y' + y'x'$$

4. **Distributive Law:** Is the following distributive law valid?

$$A \oplus (B \cdot C) = (A \oplus B) \cdot (A \oplus C)$$

No

5. **Simplification to SOP:** Reduce to a minimum sum of products:

- (a) $(A+B)(C \oplus D) + AB'$ (three terms)
 $= AC'D + AC'D' + BCD + BCD' + AB'$
 $= BCD' + BC'D + AB'$
- (b) $(W \oplus XY) + YZ + WYZ$ (four terms)
 $= WX'Y' + W'XY + YZ + WYZ$
 $= WX'Y' + YZ + W'XY + WYZ$

6. **Algebraic Simplification:** Simplify algebraically to a minimum sum of products:

$$(X+Y'+Z+W')(X+Y'+V'+W) = (X+Y')(Z+W')(V'+W)$$

$$= (X+Y')(ZV' + ZW' + WV' + W)$$

$$= XZV' + XZW' + XV' + XW' + Y'ZV' + Y'ZW' + Y'V' + Y'W$$

$$= XZW + XW'V' + Y'ZW + Y'W'V'$$

7. **Prove Validity:** Prove algebraically that the following equation is valid:

$$ABC + A'C'D' + A'BD' + ACD = (A' + C)(A + D')(B + C' + D)$$

$$(A'+C)(A+D')(B+C'+D) = (AC+A'D')(B+C'+D) = ABC + ACD + A'D'B + A'C'D'$$

8. **Expression Simplification:** Simplify each of the following expressions:

- (a) $ABCD' + A'B'CD + CD'$
 $= CD'(AB + A'B' + 1) = CD'(A+B'+1) = CD'$
- (b) $(A+B')(A'+B+D)(B'+C+D')$
 $= B' + ACD$
- (c) $[A'B' + A(B'+D)](B'+C+D')$
 $= (A'B' + AB' + AD)(B'+C+D')$
 $= (B'(A+A') + AD)(B'+C+D')$
 $= (B' + AD)(B'+C+D')$
 $= B' + ACD$

$$\begin{aligned}
 a) & P'Q'R + PQR' + PQR \\
 &= P'Q'R + PQ(R'+R) \\
 &= P'Q'R + PQ \\
 &= (P'Q'R + P)(P'Q'R + Q) \quad \rightarrow = (Q+P)(R+P)(P'+Q)(R+Q)_{\#}
 \end{aligned}$$

9. Factoring to POS: Factor to obtain a product of sums:

$$\begin{aligned}
 (a) & P'Q'R + PQR' + PQR \\
 &= (Q+P)(R+P)(P'+Q)(R+Q) \\
 (b) & PQ + P'Q' + Q'RS \\
 &= (Q'+R)(Q'+S) \\
 &= QP + P'Q' + Q'RS \\
 &= Q'P + QRS + P'Q' \\
 &= Q'(P+RS) + P'Q' \\
 &= Q'(P+P')(P+RS) + P'Q' \\
 &= Q'(P+P')(P+RS) + P'Q' \\
 &= Q'(P+P')(P+RS) + P'Q'
 \end{aligned}$$

10. Multiplying to SOP: Multiply out to obtain a sum of products:

$$\begin{aligned}
 (a) & (P'+R'+S)(P'+R)(Q+R'+S') \\
 &= P'R' + S'P' + SQR \\
 (b) & (P+Q)(P'+S)(P'+R+S) \\
 &= PS + QP' + QS
 \end{aligned}$$

$$\begin{aligned}
 b) & (P+Q)(P'+S)(P'+R+S) \\
 &= (P+Q)(P'+S) \\
 &= PS + QP' + QS
 \end{aligned}$$

$$\begin{aligned}
 a) & (P'+R'+S)(P'R)(Q+R'+S') \\
 &= [S(Q+R') + S'(P'+R)'](P'+R) \\
 &= (SQ + SR' + S'P' + S'R')(P'+R) \\
 &= (SQ + S'P' + (S+R')R')(P'+R) \\
 &= (SQ + S'P' + R')(P'+R) \\
 &= P'R' + R(SQ + S'P') \\
 &= P'R' + SQR + S'P'R \\
 &= P'(R' + S'P'R) + SQR \\
 &= P'(R' + S'P') + SQR \\
 &= P'R' + S'P' + SQR
 \end{aligned}$$